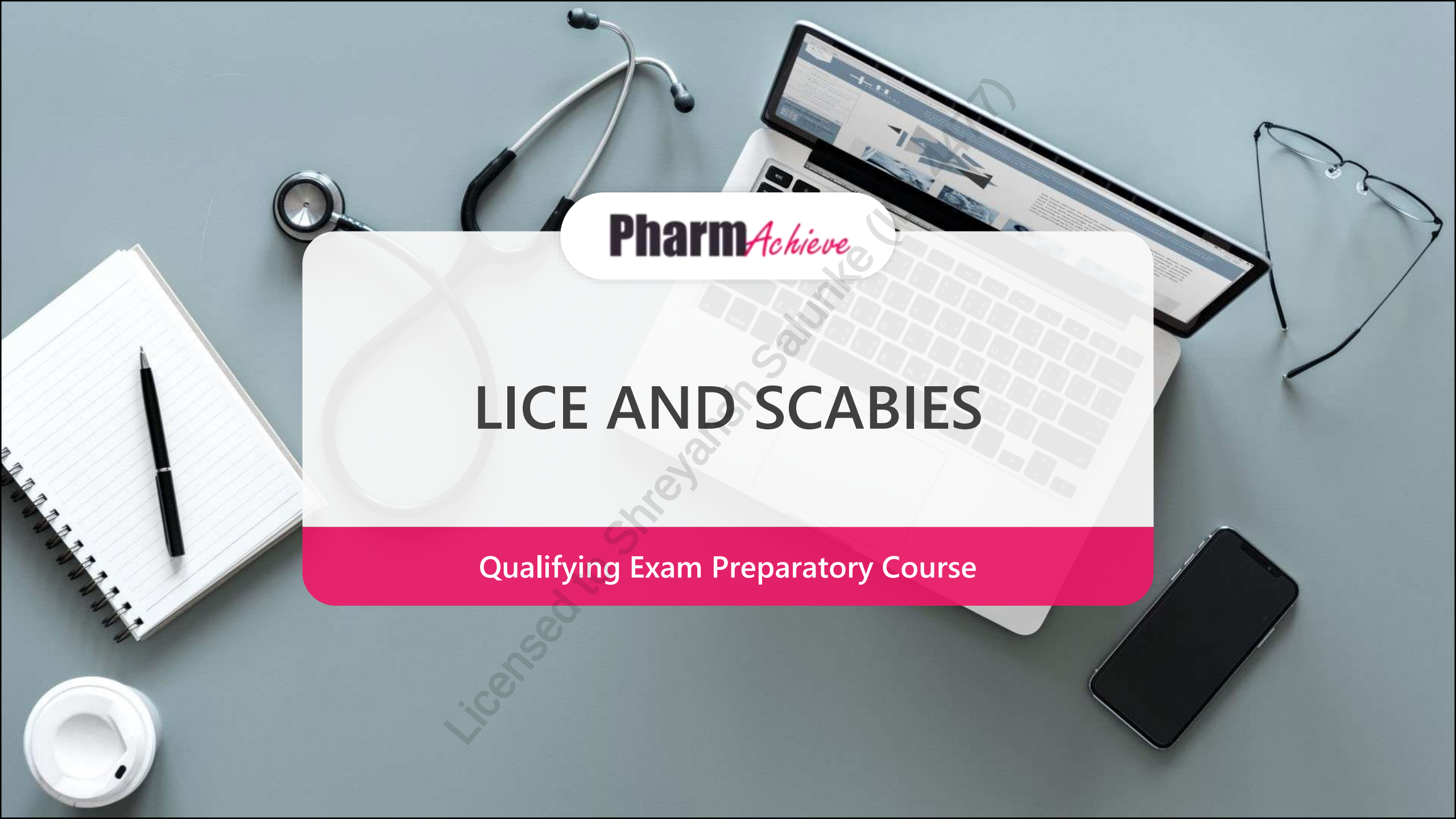




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LICE AND SCABIES

Qualifying Exam Preparatory Course

COVERED IN THIS PRESENTATION...

Ethical, Legal and
Professional
Responsibilities

Patient Care

Product
Distribution

Practice Setting

Health Promotion

Knowledge and
Research
Application

Communication
and Education

Intra and Inter-
Professional
Collaboration

Quality and Safety

LEGEND

CI Contraindication

OTC Over-The-Counter

PO Per Os (Oral)

TMP-SMX Trimethoprim-Sulfamethoxazole

Licensed to Shreyansh Salunke (ID: 4477)

OBJECTIVES

- 1 Describe characteristics of lice and scabies infections
- 2 Discuss non-pharmacological treatment & transmission prevention
- 3 Discuss therapeutic options for eradication of lice and scabies
- 4 Outline key patient counseling points for selected treatments



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LICE

Qualifying Exam Preparatory Course

DEFINITIONS

Lice are ectoparasites that commonly infect humans

- Attach to hair and feed off blood
- Pruritus caused by allergic reaction to lice saliva

There are three types of lice that infect humans and they infect different parts of the body

- Head lice is caused by *Pediculus humanus capitis*
- Body lice is caused by *Pediculus humanus corporis*
- Pubic lice ("crabs") is caused by *Phthirus pubis*

Female louse lay about 7-10 eggs (nits) per day – attach firmly at hair shaft

Nits hatch in 8-10 days releasing nymphs

Nymphs mature into adult lice in another 8-15 days

Life span of lice is about 30 days on the head

Lice can live about 1-2 days without a host

DIFFERENCES BETWEEN TYPES OF LICE

	Head Lice	Body Lice	Pubic Lice
Affected Area	Scalp and neck	Trunk	Mainly pubic area Less commonly armpits, eyebrows, facial hair
Spread By	Direct head-to-head contact, sharing personal items (e.g. combs, hats) *Human lice do not transmit via pets and animals	Clothing (mainly along the seams) and bedding, shared items, poor hygiene	Sexual contact, bed linens and towel
Epidemiology	School-aged children	People with poor hygiene, commonly in crowded areas	People who are sexually active
Louse	Eggs (nits) firmly attach to base of hair, survive 2 days away from host	Visits host to feed, survive 3 days away from host	Small and difficult to see, often seen after they feed (filled with blood)

CLINICAL PRESENTATION

- Patients may be an asymptomatic host

Common signs and symptoms include

- Pruritus
- Visualization of lice and nits
- Papules

Intense scratching may cause a secondary infection, which results in

- Pustules
- Cellulitis
- Fever
- Enlarged nuchal and cervical lymph nodes

CLINICAL PRESENTATION AND MANAGEMENT OF LICE

	Head Lice	Body Lice	Pubic Lice
Clinical features	Pruritic scalp with red papules around ears, face and neck	Pruritus and skin reactions (typically in flank, axillae, around waist and neck), lives in seams of clothing; not on body	Pruritus in anogenital area, facial hair (including eyelashes), rarely the scalp
Diagnosis	Detection of lice, eggs, or nits close to scalp	Detection of lice, eggs, or nits in seams of clothing	Detection of lice, eggs, or nits in pubic hair
Differential diagnosis	Dandruff, hair casts, hairspray droplets	Seborrheic dermatitis, flea/insect bites, eczema, folliculitis	Seborrheic dermatitis, folliculitis, dermatophytosis
Treatment	Non-Pharm and topical pediculicide * Only treat if live lice are present * Pets should not be treated	Non-pharm generally adequate: Bathe thoroughly, wash clothes and linens on hot cycle	Non-pharm and topical pediculicide



*Treatment should not be initiated unless there is a clear diagnosis (live lice visualized)

GOALS OF THERAPY

- ✓ Treat the signs and symptoms
- ✓ Eliminate the parasites and prevent recurrence
- ✓ Prevent complications (secondary bacterial infections)
- ✓ Prevent transferring and infecting others

NON-PHARMACOLOGICAL THERAPY

- Examine close human contacts for infection, treat if needed
- Avoid sharing belongings and direct physical contact to prevent transmission
- Physically remove nits with nit comb after pharmacologic treatment *OR* wrap hair in vinegar-soaked towel for 30-60 min before wet combing
 - Poor efficacy with wet combing alone (comb wet hair with thin, fine-toothed comb every 3-4 days)
- Disinfect combs and brushes with Lysol 2% (1 hour), rubbing alcohol (10-20 min), or hot water (5-10 min)
- Wash all exposed bedding, towels, personal articles & clothing in hot cycle
- For items that cannot be washed, seal in plastic bag x 2 weeks
- Vacuum areas (e.g. furniture, mattresses, flooring) that came into contact with the infected person(s)

*Hot air and electric combs have inferior evidence for efficacy

PHARMACOLOGICAL THERAPIES FOR LICE

	Permethrin 1% Nix® Crème Rinse, Kwellada-P® Creme Rinse	Pyrethrins/piperonyl butoxide R&C Shampoo®	Isopropyl myristate/ cyclomethicone Resultz (50%/50%)®	Dimeticone 50% NYDA®
Mechanism of Action	Neurotoxic insecticide of the pyrethrin family	Neurotoxic insecticide of the pyrethrin family	Disrupts wax layer of louse exoskeleton, physically kills via dehydration	Silicone based oils that coats louse and physically kills via suffocation
Place in Therapy	1 st line; used in both head and pubic lice	1 st line; used in both head and pubic lice	2 nd line	2 nd line
Side Effects	Local irritation	Local irritation	Local irritation	Local irritation
Special Considerations	CI in children < 2-month-old; compatible in pregnancy and breastfeeding	CI in children < 2-month-old; compatible in pregnancy and breastfeeding	Not recommended in children < 4 years old, pregnancy or breastfeeding	Not recommended in children < 2 years old, pregnancy or breastfeeding
Comments	Treatment failure may be due to resistance; repeat treatment in 7 days	Treatment failure may be due to resistance; repeat treatment in 7 days	Unlikely to develop resistance; repeat treatment in 7 days	Unlikely to develop resistance; repeat treatment in 8-10 days

PHARMACOLOGICAL THERAPIES FOR LICE

- Controversy re: allergies
- Pyrethrin is made from natural chrysanthemum extracts
 - Possible true allergic reaction in patients with ragweed allergies due to cross-sensitivity
 - Contraindicated in patients with chrysanthemum allergy due to extracts used to make drug
- Permethrin is a synthetic pyrethroid
 - Canadian Pediatric Society states that it “does not cause allergic reactions”
 - However, drug monograph states contraindication in patients with chrysanthemum hypersensitivity
- Bottom line: Pyrethrin is CI in chrysanthemum allergy, possibly ragweed allergy due to cross-sensitivity
 - Permethrin is likely safe in chrysanthemum/ragweed allergy

APPLYING TOPICAL PEDICULICIDE

Permethrin: 1st line therapy for head lice and pubic lice

Applying head lice treatment

- Wash hair with conditioner-free shampoo and towel dry
- Apply treatment to hair and scalp, ensuring hair is thoroughly soaked
 - ½ to 1 bottle for adults and children with long hair
- Leave treatment on for 10 minutes then rinse
- Comb out and remove lice and nits
- *May* repeat treatment in 1 week if live lice are observed (re-treatment after 1 week is recommended though since pediculicides are not reliably 100% ovicidal)

Applying pubic lice treatment

- Apply and saturate pubic hair
- Leave treatment on for 10 minutes then rinse
- Repeat treatment in 7-9 days



APPLYING TOPICAL PEDICULICIDE

Pyrethrins/piperonyl butoxide: 1st line therapy for head lice and pubic lice

Head lice treatment (same steps for pubic lice)

- Apply treatment (lice shampoo) to dry hair and scalp
- Massage until hair is completely saturated
- Leave treatment on for 10 minutes then add water
- Lather then rinse over sink
- Comb out the lice and nits
- Repeat treatment in 7 days



APPLYING TOPICAL PEDICULICIDE

Isopropyl myristate/cyclomethicone

- Apply treatment to dry hair and scalp
- Massage until hair is completely saturated
- Leave treatment on for 10 minutes
- Rinse with warm water
- Comb out the lice and nits
- Repeat treatment in 7 days



APPLYING TOPICAL PEDICULICIDE

Dimeticone 50%

- Spray onto dry hair and scalp
- Massage until hair is completely saturated
- Leave on for 30 minutes
- Comb out lice and nits
- Leave solution to dry on hair for at least 8 hours
- Wash hair
- Repeat in 8-10 days



PHARMACOLOGICAL THERAPIES FOR LICE

In cases of head lice treatment failure

- Switch to agent of different pharmacologic class
- **Permethrin 5% cream** applied to scalp and left on overnight
- **Oral TMP-SMX with permethrin 1%** applied to scalp and left on for 24h (adults) (limited data on safety and efficacy)
- Oral **ivermectin** 200 mcg/kg repeated in 7-10 days or 400 mcg/kg repeated in 7 days

Unconventional treatments

- Cetaphil cleanser (Nuvo® lotion) applied to scalp to dry in place shown to be effective in one uncontrolled study
 - No substantial evidence for hot air, hair styling gels, mayonnaise, vinegar, isopropyl alcohol, olive oil, melted butter, petroleum jelly, acetomicellar complex, or tea tree oil

Itching alone is NOT a sign of treatment failure (side effect of pediculicides)

- Can be managed with oral antihistamines or topical corticosteroids
 - E.g. Cetirizine (safe for use in pregnancy and breastfeeding)

MONITORING PARAMETERS

EFFICACY ENDPOINTS			
Parameter	Frequency of monitoring	Degree of Change	Timeframe
Live lice	Daily	Eradicated	7-10 days
Presence of nits		Eradicated	7-14 days
Pruritus		None	4-6 weeks
Secondary infection (fever, cellulitis, pustules, enlarged lymph nodes)		Prevent	Duration of therapy

SAFETY ENDPOINTS			
Parameter	Frequency of monitoring	Degree of Change	Timeframe
Adverse reactions and Allergic reactions	Day(s) of application	None	Duration of therapy

TAKEAWAYS

- Lice are ectoparasites that can infest the scalp, body, or pubic region
- Common **symptoms include pruritus, visible lice and nits (eggs), and papules**
- Non-pharmacological management
 - Physically remove nits with a nit comb after pharmacologic treatment
 - Wash bedding, towels, and personal items in a hot cycle, or seal in a plastic bag for 2 weeks
- Pharmacological management
 - First-line therapies: **permethrin 1%, pyrethrins/piperonyl butoxide** (contraindicated in chrysanthemum allergy)
 - Contraindicated in children <2 months old
 - Second-line therapies: **isopropyl myristate/cyclomethicone** (not recommended in < 4 years old), **dimeticone 50%** (not recommended in < 2 years old)
- Repeat treatment if suspected failure (indicated by the presence of live lice) in **7-10 days**
- **Persistent treatment failure:** switch to agent of a different class, permethrin 5%, TMP-SMX PO with permethrin 1%, ivermectin PO

CASE STUDY

JN is a 9-year-old female who is brought to the pharmacy by her father because she has been scratching her scalp for several days. On physical examination, live lice and nits are observed on JN's scalp. JN has a history of asthma and seasonal allergic rhinitis, for which she takes salbutamol 100 mcg 2 inhalations Q4-6H PRN, fluticasone propionate 100 mcg 1 inhalation BID, and loratadine 10 mg PO daily during allergy season. JN lives at home with her two siblings, one brother (aged 6) and one sister (aged 12). JN's father tells you that JN was sent home from school when her teacher noticed the live lice a few days ago, and that she has not been back since. He is also worried that he will require treatment along with the whole household. He explains that none of his kids have experienced head lice to date and is wondering if JN's long hair contributed to its development. JN has an allergy to ragweed that results in localized urticaria.

CASE STUDY - QUESTIONS

1. Which of the following medications is most appropriate to initiate in JN?
 - a) Permethrin 1% cream rinse
 - b) Pyrethrins/piperonyl butoxide shampoo
 - c) Dimeticone 50% spray
 - d) Permethrin 5% cream
2. Which of the following non-pharmacologic options is LEAST appropriate to recommend?
 - a) Vacuum all areas that JN has come into contact with
 - b) Seal items that cannot be washed in a plastic bag for 2 weeks
 - c) Wet combing may be used in conjunction with pharmacological management for improved results
 - d) Electric combs demonstrate better outcomes compared with traditional nit combs
3. What is the most appropriate statement to tell JN's father regarding the treatment of household contacts?
 - a) Household contacts should be treated prophylactically
 - b) Treat household contacts if there is evidence of live lice
 - c) Household contacts may receive treatment if they have an itchy scalp
 - d) Household contacts can receive treatment one day after JN is treated
4. Which of the following statements is the most accurate to tell JN's father regarding JN's hair length?
 - a) JN's hair should be cut following treatment
 - b) Hair length does not impact susceptibility
 - c) Prophylactic treatment is recommended for children with long hair
 - d) Long hair only increases severity of the infection, not susceptibility



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SCABIES

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DEFINITION

- Scabies are a parasitic skin infection caused by mites called *Sarcoptes scabiei* var. *hominis*
- *Sarcoptes scabiei* life cycle:
 - After fertilization, female mites burrow into the skin and lay 2-3 eggs per day for entire lifespan of 4-6 weeks
 - In 3-4 days, larvae hatch and head for skin surface to find a mate
 - Mature into adult mites in 14-17 days
 - Process repeats
 - Can survive away from host for 2-36 hours at room temperature, and up to 19 days in cool, humid environments
- Hypersensitivity (type IV) to the mites, eggs, and feces causes intense pruritus that worsens at night

RISK FACTORS & EPIDEMIOLOGY

- Scabies primarily spreads through direct contact
 - Commonly spread through sexual contact
- Uncommon transmission through bedding and furniture
 - Unless high parasite load
- Prevalence increases in crowded conditions
- Institutional environments (nursing homes, prisons) may have scabies outbreaks
- Human scabies is not transmitted by animals
 - A different scabies mite, *Sarcoptes scabiei var canis*, which does not survive and reproduce on humans, may cause an infestation known as “mange” in animals such as cats or dogs

CLINICAL PRESENTATION OF SCABIES

- Common signs and symptoms include:
 - Intense pruritus that worsens at night
 - Burrows on hands and flexor surfaces
 - White/grey linear or wavy lines beneath surface of skin
 - Papules (commonly on trunk)
 - Secondary infections
 - “Crusted scabies” in immunocompromised hosts are more contagious with a high secondary infection rate
 - Atypical presentation in children on hands, feet, scalp, body folds
- Patients may be asymptomatic for up to 6 weeks after infestation
 - Due to delayed type IV hypersensitivity
 - No delay in symptoms in re-infestation due to prior sensitization

Distribution:

- Webs of the fingers, wrists, axillae, knees
- Male genitalia and skin under breasts
- Skin folds around belt line, navel, buttocks, thighs

DIAGNOSIS AND TREATMENT

- Physicians should make diagnosis before starting scabies treatment
 - May be based off signs and symptoms
 - May require skin scraping test
- Differential diagnoses
 - Flea/insect bites, atopic/seborrheic dermatitis, impetigo
- Treatment
 - Non-pharmacological measures (wash linens and clothing in hot cycle)
 - Avoid close contact with others until completion of treatment/follow-up
 - All contacts (sexual, close personal and household contacts) within the preceding month require treatment, even if asymptomatic

GOALS OF THERAPY

- ✓ Treat the signs and symptoms
- ✓ Eliminate the parasites
- ✓ Prevent complications (secondary bacterial infections)
- ✓ Prevent transferring and infecting others

NON-PHARMACOLOGICAL THERAPY

- Avoid sharing belongings and direct physical contact to prevent transmission
- Wash linens and clothes in hot cycle
- Seal items that cannot be washed
 - At least 3-7 days for scabies
- Vacuum areas that came in contact with the infected person(s)
- Children can return to school day after treatment is completed
- Trim fingernails to avoid injury from excessive scratching

PHARMACOLOGICAL THERAPY: SCABIES

	Permethrin 5%	Sulfur 5-10%
Place in Therapy	1 st line Drug of choice in pregnancy and breastfeeding	2 nd line Safe in pregnancy and children < 2 months Compounded in petrolatum base
Side Effects	Local irritation	Local irritation, malodourous, stains clothing
CI	Infants < 2 months of age	None indicated
Application <i>*Refer to next slide for details</i>	Apply to all skin areas and leave on for 8-14 hours, then wash off	Apply to all skin areas at bedtime for 3 consecutive days Leave on for 24 hours, wash off prior to next application



Crotamiton 10% cream was a second-line agent that is no longer available in Canada

APPLYING SCABICIDE

- Apply to clean dry skin
- Massage thin layer of cream/lotion into all skin areas from the neck down to the toes (including fingernails, waist, genitalia)
 - In young children, treat entire head and neck if possible
- Leave on for a certain number of hours (depends on treatment) then wash off
- May repeat treatment in 1 week if mites are still active
- Pruritus may persist for 1-2 weeks after successful treatment

MONITORING PARAMETERS

EFFICACY ENDPOINTS			
Parameter	Frequency of monitoring	Degree of Change	Timeframe
Burrows	Daily	Eradicated	7-14 days
Papules			7-14 days
Pruritus			4 weeks
Fever, cellulitis, enlarged lymph nodes (secondary bacterial infection)		Prevent	Duration of therapy

SAFETY ENDPOINTS			
Parameter	Frequency of monitoring	Degree of Change	Timeframe
Adverse effects/Allergic reactions	Day of application	None	Duration of therapy

TAKEAWAYS

- All the topical products are external use only
 - Avoid eyes and mucous membranes
 - May cause local irritation at site of application
- Treatments should be repeated in 7 days if new burrows or papules are observed (pruritus alone is not a sign of treatment failure)
- Antihistamines or topical corticosteroids may be used to control pruritus
 - E.g. betamethasone valerate 0.1% cream
- Refer:
 - Secondary infection or suspected crusted scabies
 - If parasites are not eradicated after OTC treatment (2 applications separated by a week)
 - May require prescription therapy, e.g. oral ivermectin 200 mcg/kg (no longer requires Special Access Programme approval through Health Canada)

CASE STUDY

TD is a 28-year-old female who presents to the pharmacy and requests to speak with the pharmacist in the private counselling room. She tells the pharmacist that she thinks that she has scabies as she is experiencing intense pruritus in the folds of her skin that worsens at night, and grey lines that look like burrows on her arms. TD has never had scabies before, however, a friend of hers has had it in the past. She wants to know whether showering more frequently will help her to avoid the development of scabies in the future and informs you that she currently showers every morning. TD shares an apartment with one roommate, and she has multiple sexual partners. Her past medical history is significant for hypothyroidism and mild papulopustular acne. Her current medications include levothyroxine 75 mcg PO daily, and benzoyl peroxide 5%/clindamycin 1% gel applied BID, Alesse® 28 (levonorgestrel 100 mcg/ethinyl estradiol 20 mcg) PO daily, and a multivitamin supplement 1 tablet PO daily.

CASE STUDY - QUESTIONS

1. Which of the following counselling points is most appropriate to review with TD?
 - a) Avoid cutting fingernails for the duration of treatment
 - b) Sexual and household contacts within the preceding month require treatment
 - c) Items that cannot be washed should be sealed in a plastic bag for at least 2 weeks
 - d) Personal belongings may be shared between members of the same household
2. Which of the following statements on the application of permethrin 5% cream is FALSE?
 - a) Apply to the skin after showering
 - b) Remove the cream by washing with a shower or bath after approximately 8-14 hours
 - c) A single application will usually suffice to clear the infestation
 - d) A second application can be given after 7-10 days if live mites or new lesions appear
3. Which of the following statements is most accurate regarding the transmission of scabies?
 - a) Transmission is related to direct contact, not personal hygiene
 - b) Increasing the number of showers per day is likely to prevent scabies in the future
 - c) Scabies is due to excessive sweating, so keeping the skin dry will prevent scabies
 - d) Scabies is commonly transmitted through bedding and furniture
4. TD returns to the pharmacy after one week and reports continued pruritus, but no new lesions or burrows. What is the most appropriate recommendation to make to TD?
 - a) Oral ivermectin
 - b) Topical mometasone ointment
 - c) Repeat application of permethrin 5% cream
 - d) Minimize body washing for the next week to avoid further irritation

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CHANGE LOG

March 2025

- Content reviewed; formatting changes made

May 2025

- Slide 24: Changed 5-7 days to 3-7 days for duration to seal items
- Slide 25: Updated sulfur concentration range in line with CDC
- Slide 28: Updated references

July 2025

- Content reviewed; no changes made

September/October 2025

- Minor formatting and grammar changes made
- Slide 3: Added legend
- Slide 12: Updated age indication of Resultz®
- Slides 19, 32: Added Monitoring Parameters
- Slide 20: Added takeaway
- Slides 21, 22, 34, 35: Added case studies and questions
- Slide 36: Updated references